

Lecture 6: Analysis of DNA by Agarose Gel Electrophoresis

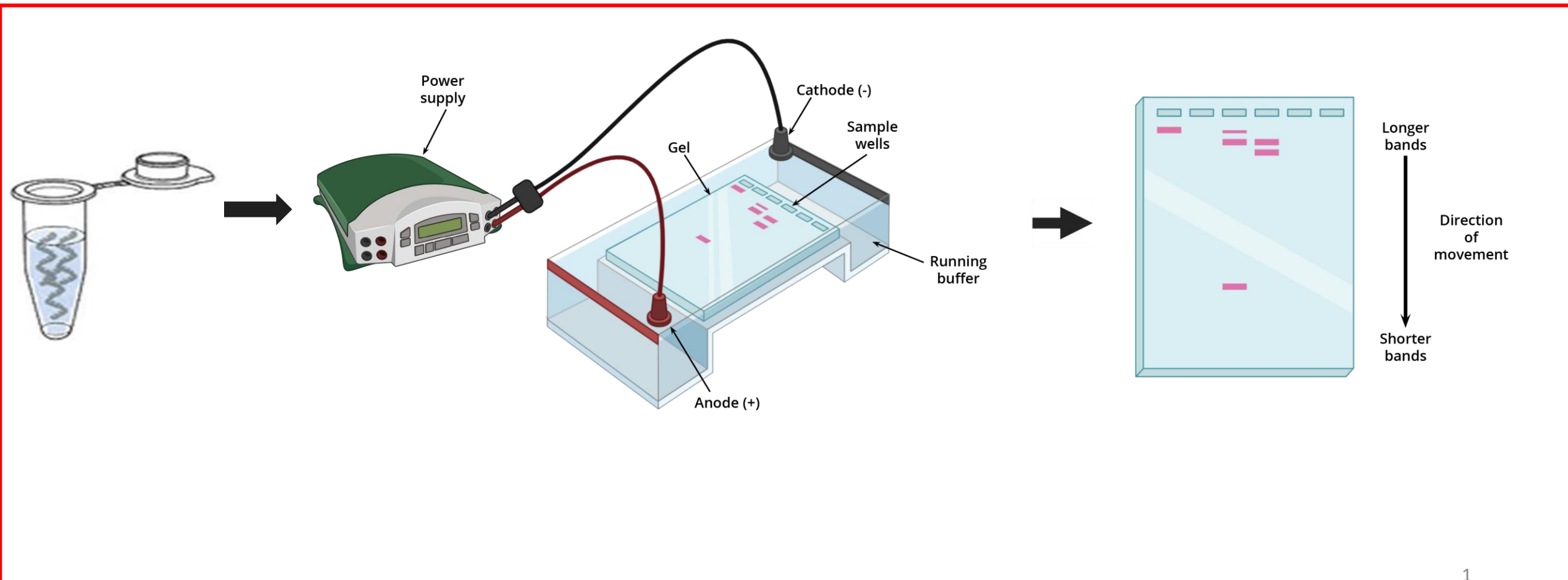
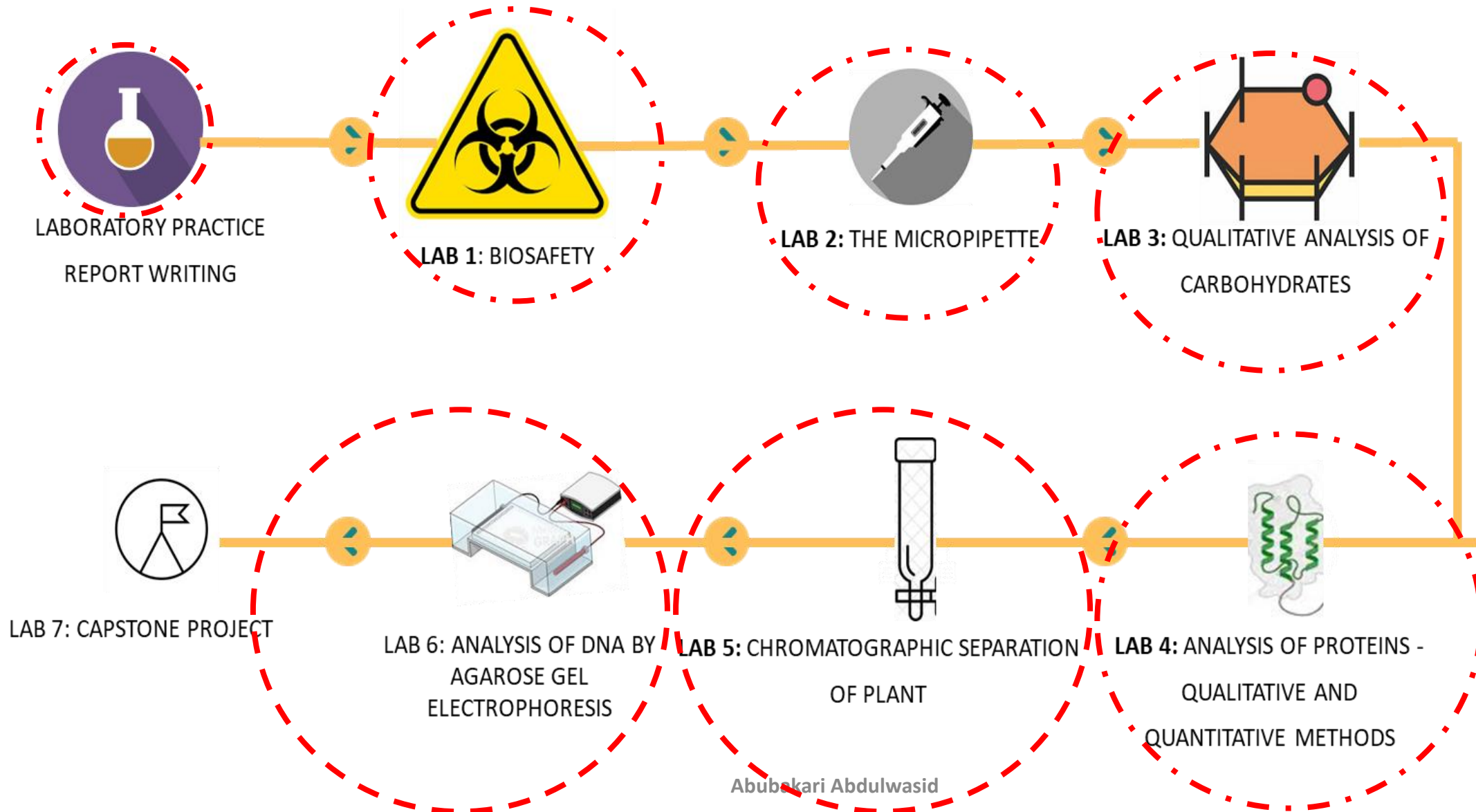


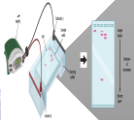
Table of content for the Course



This week Lecture



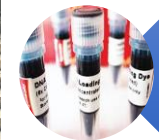
Explain the fundamental principles of separating DNA by gel electrophoresis.



Identify the key components and reagents of the system



Describe the function of agarose, buffer, loading dye, and DNA stain.



Outline the procedure for casting and running an agarose gel.



Interpret gel results by estimating DNA fragment size using a DNA ladder.



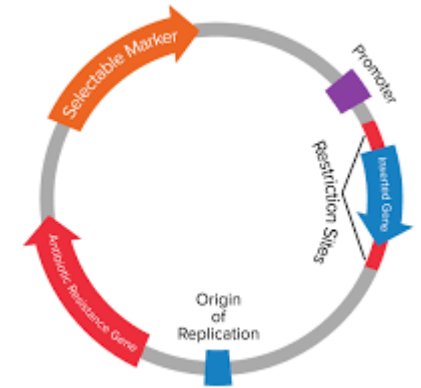
The "Big Picture": Why Separate DNA?

Why do we need this technique?

In molecular biology, our DNA samples are often complex mixtures. We always need to ask

- ❖ Did my PCR amplification work?
- ❖ What DNA fragments are in this mixture?
- ❖ Isolate a specific piece of DNA for cloning.
- ❖ Does this DNA sample match the suspect?

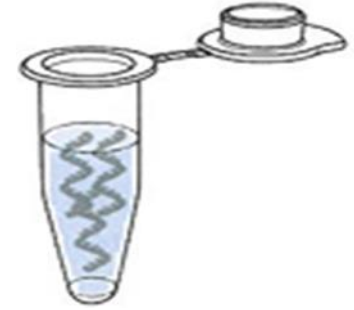
Agarose Gel Electrophoresis is our primary tool to answer these questions by sorting DNA fragments.



The Central Challenge: How Do We Sort DNA?

Imagine you have **a sample** containing **millions of DNA fragments** of different lengths. **How can you sort them?**

We use **Agarose Gel Electrophoresis**, a technique that acts like a "**molecular sieve**" to separate **DNA based on its size**.



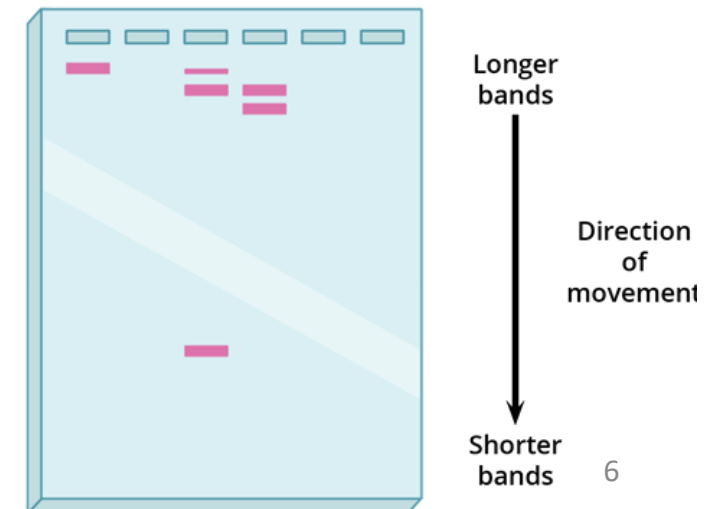
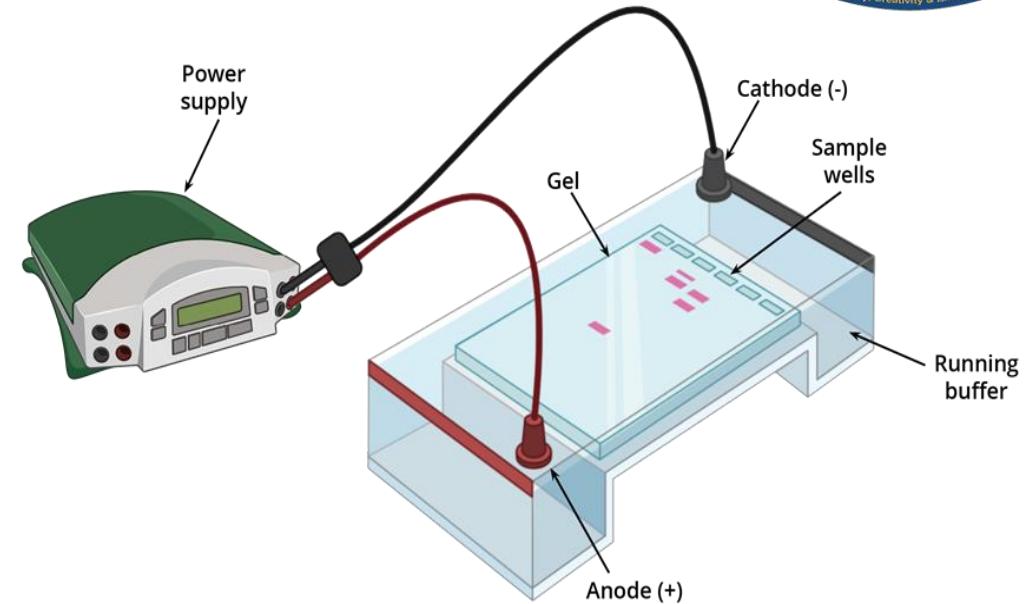
How Electrophoresis Works: The Core Idea

Using Force to Sieve Molecules

The name tells you how it works:

- ❖ **Electro-**: Uses an electric field as the driving force.
- ❖ **-phoresis**: From the Greek phoros, meaning "to carry across."

We "carry" charged molecules across a gel with an electric field. The gel acts as a molecular sieve, separating the molecules based on how easily they can move through its pores.

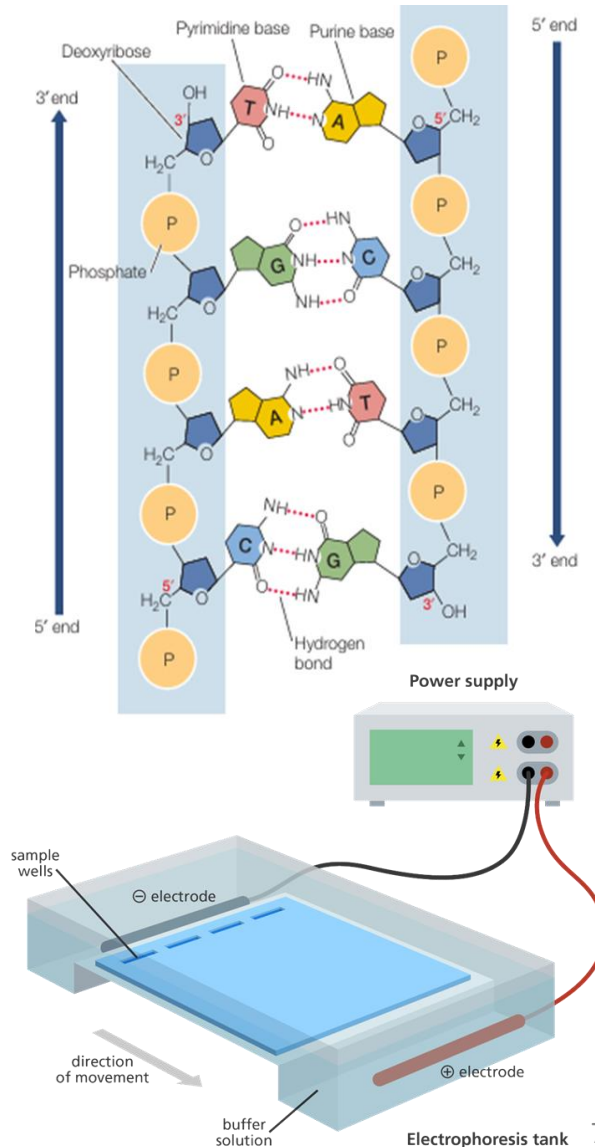


Principle 1: DNA Has a Negative Charge

The Engine of Movement

- ❖ DNA's sugar-phosphate backbone is full of **phosphate groups (PO_4^{3-})**.
- ❖ **At neutral pH**, these groups are negatively charged.
- ❖ This gives the entire DNA molecule a strong, uniform **net negative charge**.
- ❖ Therefore, when placed in an electric field, DNA will always migrate toward the positive electrode.

Mnemonic: "Run to Red"

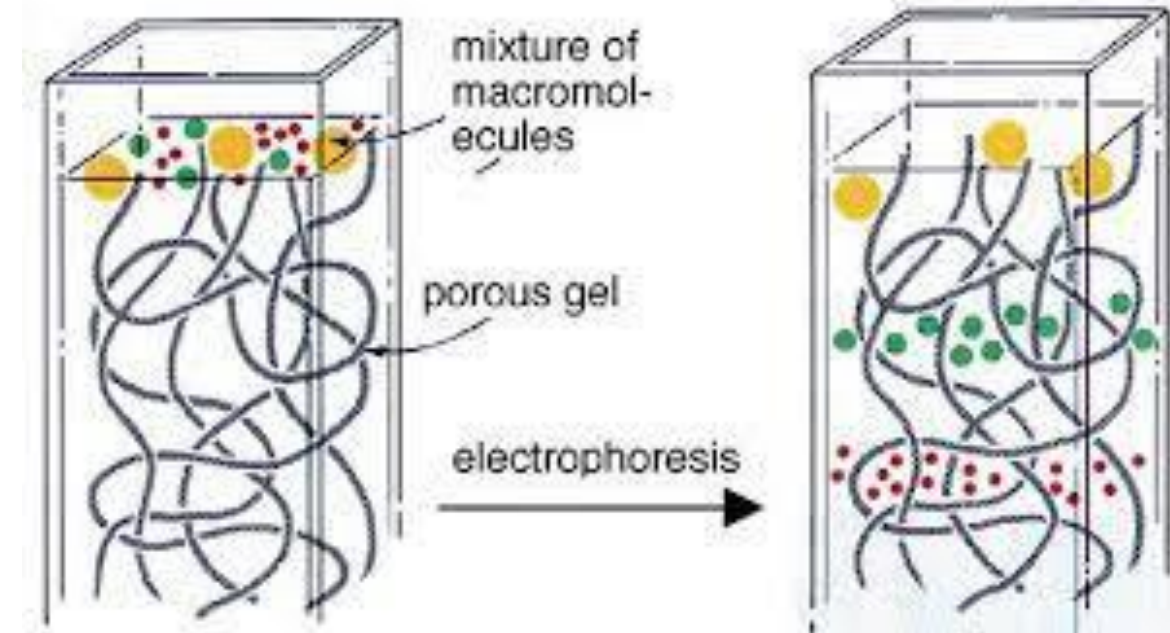


Principle 2: The Gel Separates by Size

The Gel as a Sieve: The agarose gel forms a three-dimensional matrix of pores.

- ❖ **Small DNA fragments** navigate the pores easily and travel a long distance through the gel.
- ❖ **Large DNA fragments** get tangled and are impeded by the pores, so they travel a short distance.

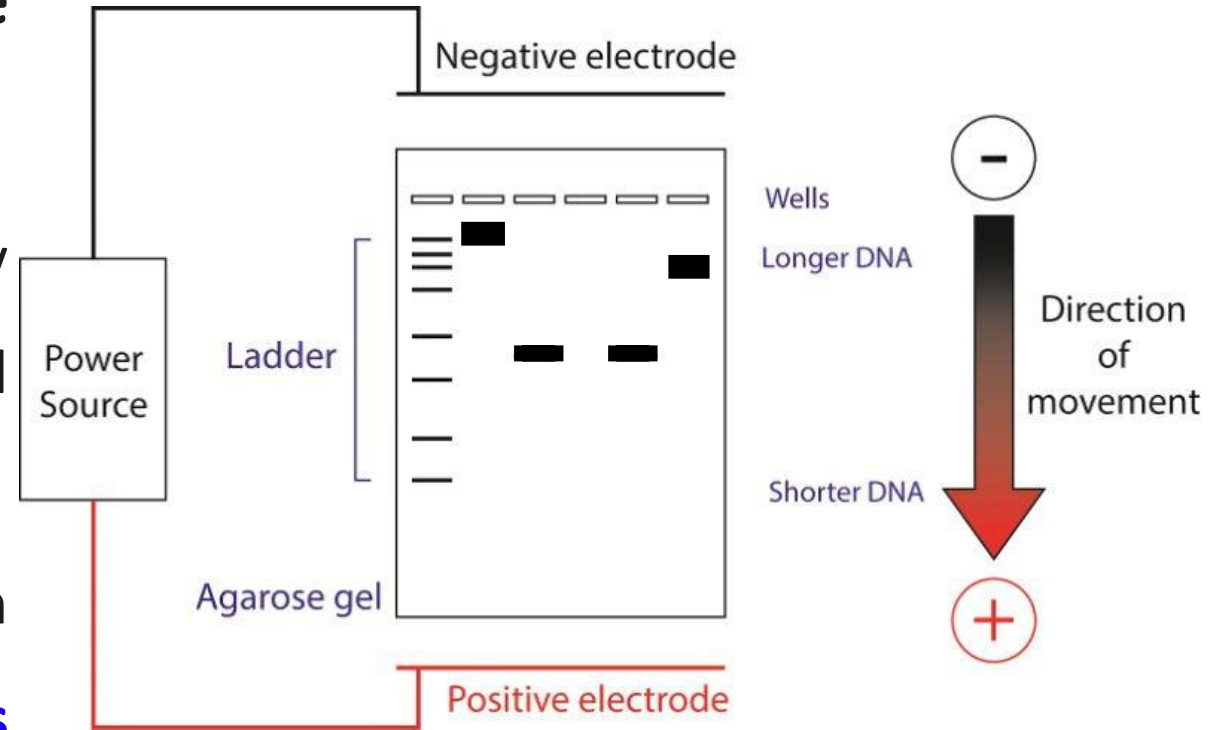
DNA is separated by size. Smaller fragments are at the bottom (far from wells), larger fragments are at the top (close to wells).



The Two Principles Together

Charge Starts the Race, Size Determines the Winner.

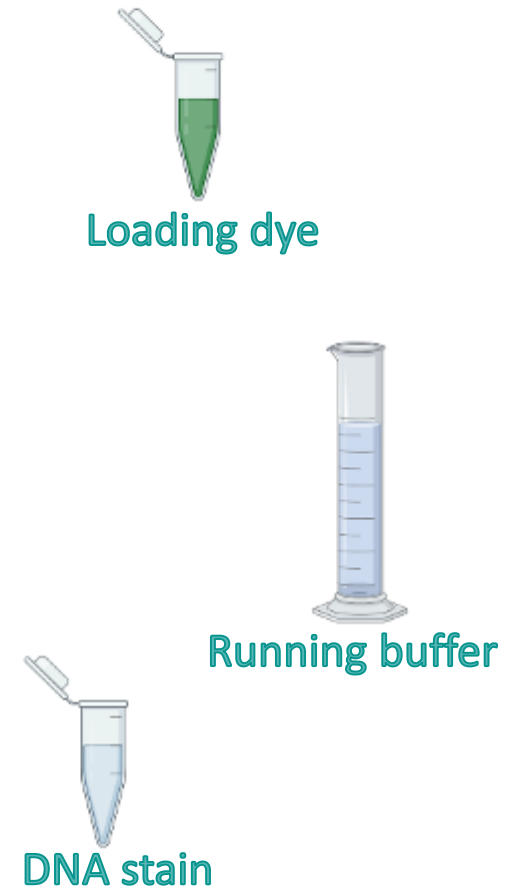
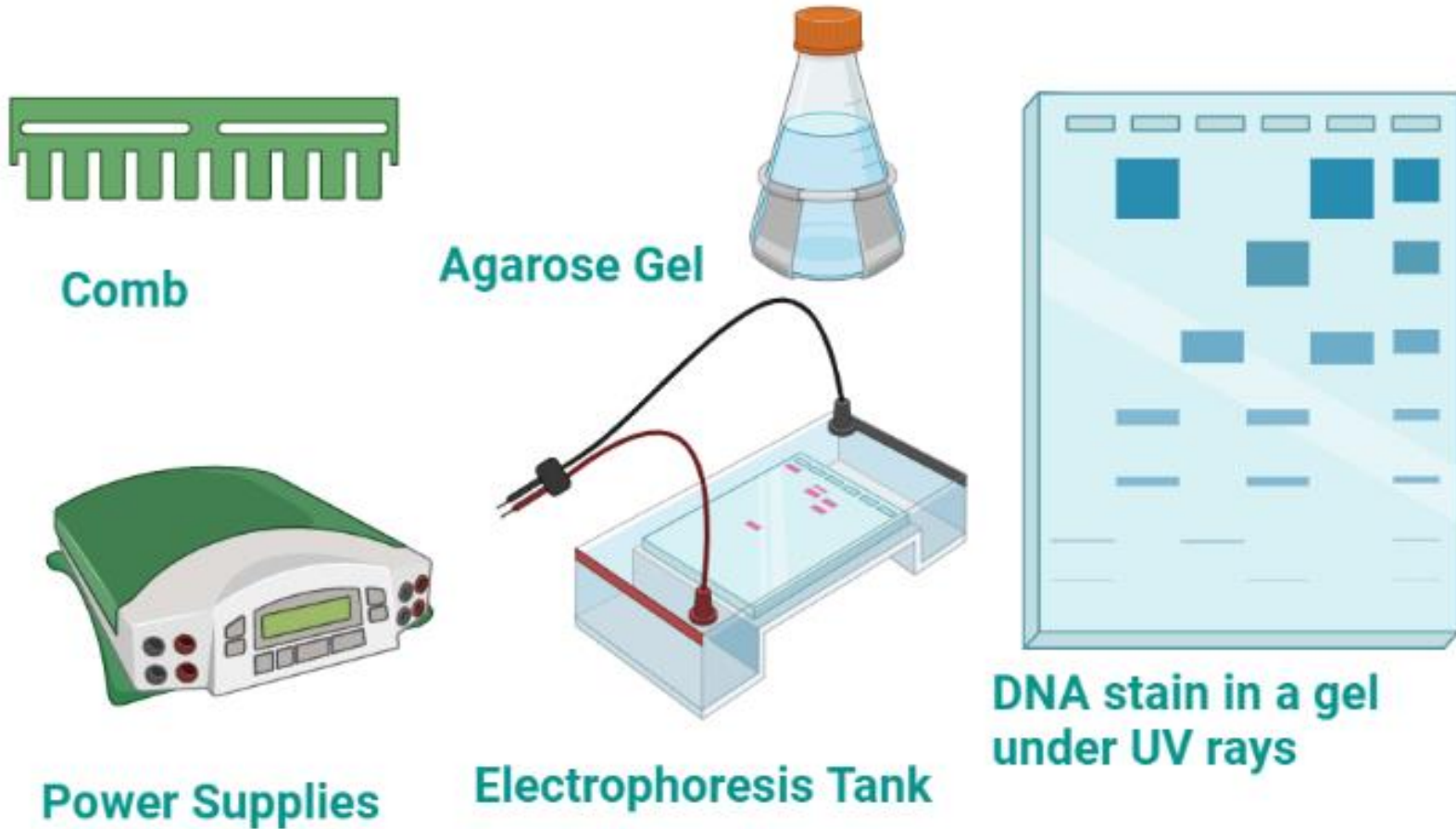
- ❖ The **electric field** makes all the negatively charged DNA fragments start moving toward the positive pole. (Everyone starts the race).
- ❖ The **agarose sieve** then slows them down differently based on their size. **Small fragments** navigate the pores easily and move fast, while **large fragments** get tangled and move slowly.



The final position of a DNA band depends almost entirely on its size.

The Toolkit: Building Our System

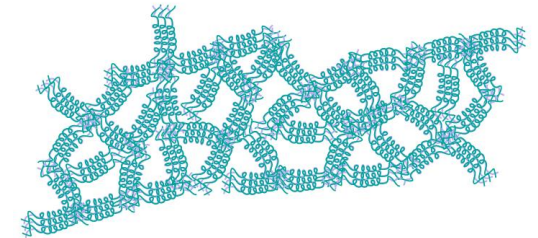
The Toolkit: Components & Reagents



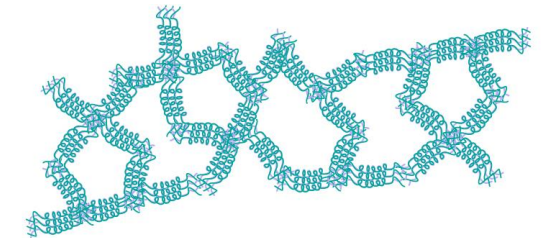
The Sieve: The Agarose Gel Matrix

The Sieve Itself

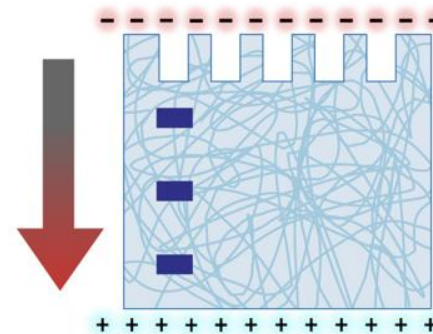
- ❖ **Agarose:** A natural polysaccharide from **seaweed**.
- ❖ Forms a porous, semi-solid gel when heated and cooled.
- ❖ **Concentration is key:**
 - ✓ **Low % gel (0.8%):** Larger pores -> Better for separating LARGE DNA.
 - ✓ **High % gel (2.0%):** Smaller pores -> Better for separating SMALL DNA.



GOOD FOR SEPARATING
SMALL DNA PIECES



GOOD FOR SEPARATING BIG DNA PIECES

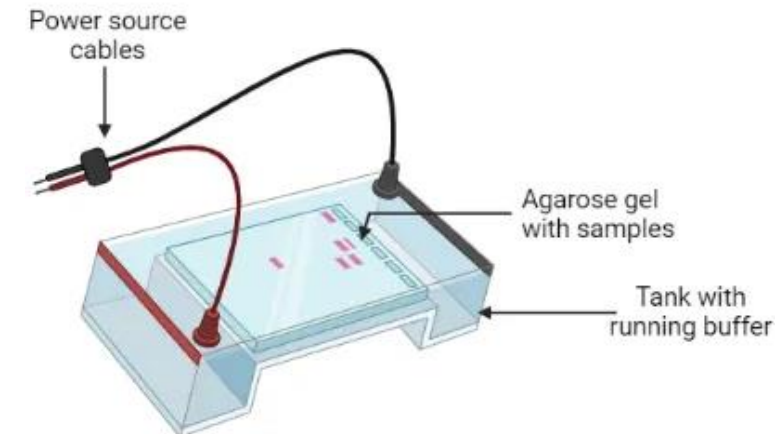
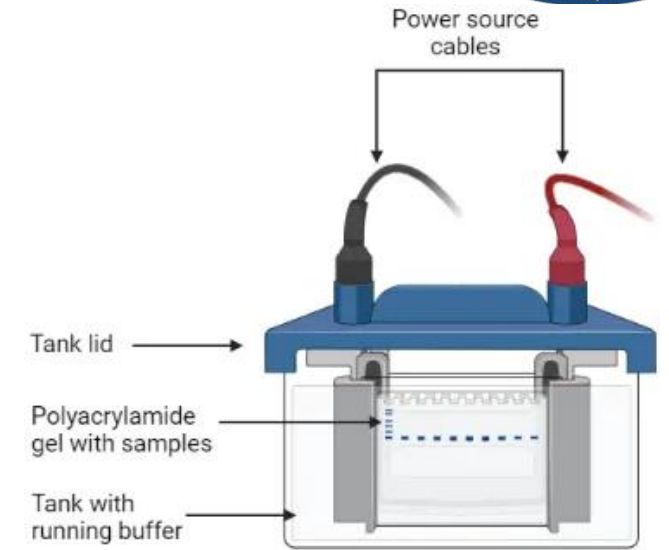


The Environment: The Buffer

More Than Just Water

We use a buffer (**Tris-Acetate-EDTA & Tris-Borate-EDTA**) for two critical reasons:

- ❖ **Conductivity:** Water alone is a poor conductor. The buffer contains ions (salts) that conduct the electric current from the power supply, through the gel, completing the circuit. Without it, the DNA won't move.
- ❖ **pH Stability:** The buffer maintains a stable pH, which ensures the phosphate groups on the DNA remain negatively charged.



The "Invisible" Problem

DNA is colorless, and the gel is transparent.

How do we **load our samples** and **see the final results**? We need **two key reagents**:

❖ **Loading Dye**: To help us see and load the sample.

❖ **DNA Stain**: To help us see the separated DNA bands after the run.



- ✓ Purple Gel Loading Dye (NEB)
- ✓ BlueJuice™ Gel Loading Buffer
- ✓ TrackIt™ Loading Buffers
- ✓ GelPilot Loading Dye (QIAGEN)
- ✓ 6X Orange DNA Loading Dye

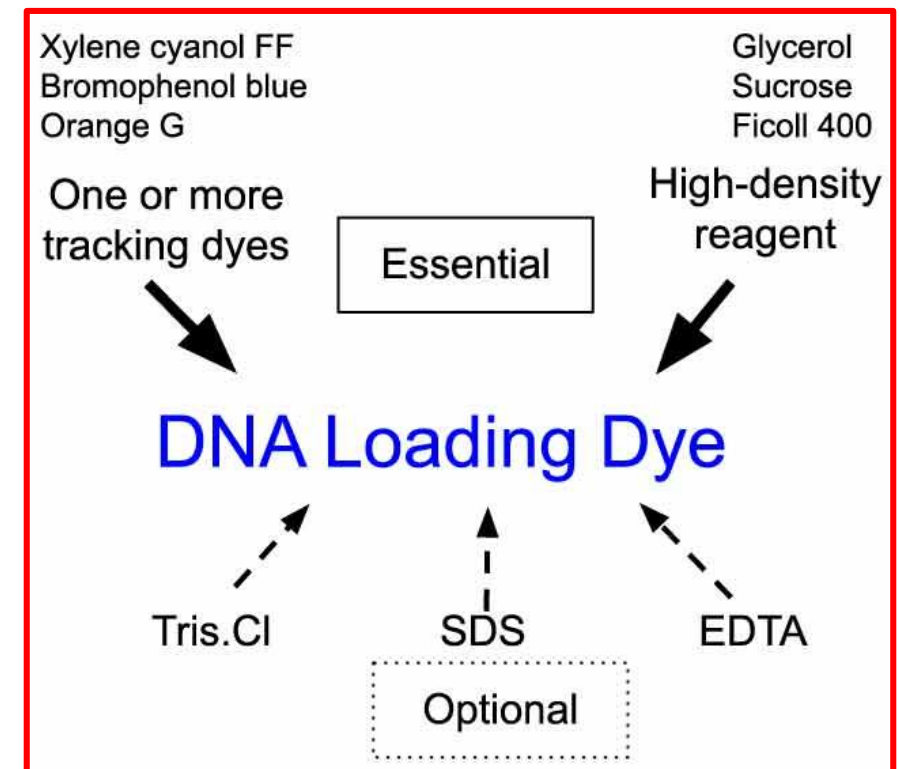
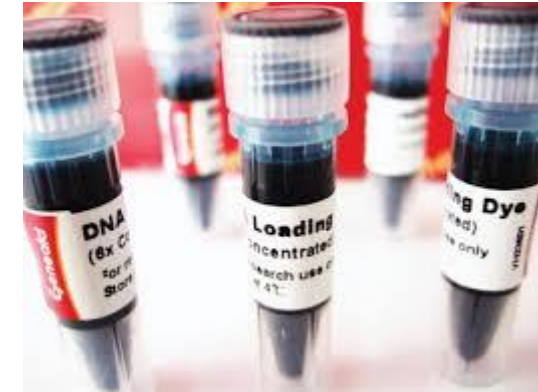


- ✓ Ethidium Bromide (EtBr)
- ✓ SYBR® Family Dyes
- ✓ GelRed™
- ✓ GelGreen™

Solution 1: Loading Dye

Two Key Functions:

- ❖ **Density:** Contains a dense substance (like glycerol or sucrose) that makes the DNA sample heavier than the buffer. This allows the sample to sink neatly into the well.
- ❖ **Tracking:** Contains one or more colored dyes (like bromophenol blue) that are also negatively charged and move through the gel, allowing us to visually monitor the progress of the electrophoresis run.



Solution 2: DNA Stain

Making the DNA Fluoresce

- ❖ We use a **DNA-binding dye** (e.g., SYBR Safe) that **intercalates (inserts itself)** into the **DNA double helix**.
- ❖ This dye is normally **invisible**.
- ❖ However, when the gel is exposed to **ultraviolet (UV) light**, the dye bound to the DNA fluoresces, revealing the location of the DNA bands.

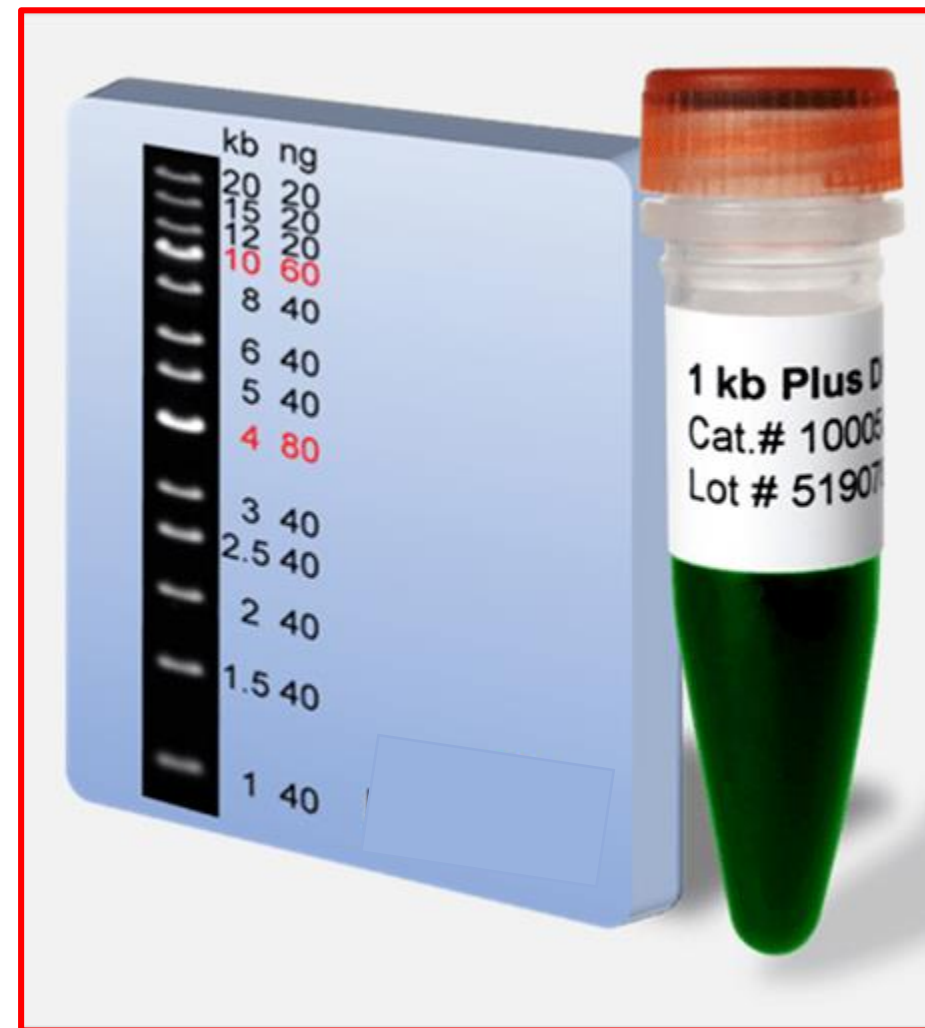


- ✓ Ethidium Bromide (EtBr)
- ✓ SYBR[®] Family Dyes
- ✓ GelRed[™]
- ✓ GelGreen[™]

The "Ruler": The DNA Ladder

Estimating the Size of Unknowns

- ❖ A DNA Ladder (or Marker) is a mixture of DNA fragments of known sizes.
- ❖ It is always run in the first lane of the gel.
- ❖ By comparing the position of our unknown bands to the bands of the ladder, we can estimate their size. It is our "molecular ruler."

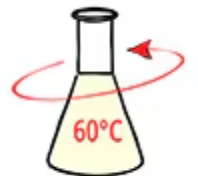
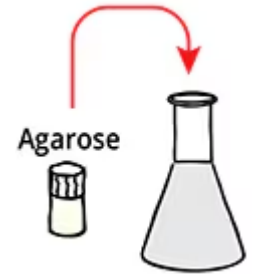
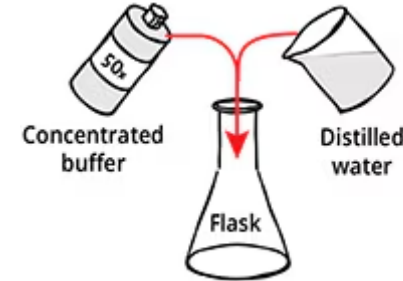


From Powder to Picture: A Step-by-Step Guide



Step 1: Preparing the Agarose Solution

- ❖ Determine the **correct amount of agarose powder** for your desired gel concentration and volume.
- ❖ Combine the **agarose powder with the correct volume of electrophoresis buffer** (e.g., TAE or TBE) in a flask.
- ❖ Heat the mixture in **a microwave or on a hot plate**, swirling carefully, until the solution becomes perfectly clear. All agarose grains must be fully dissolved for the gel to form correctly.

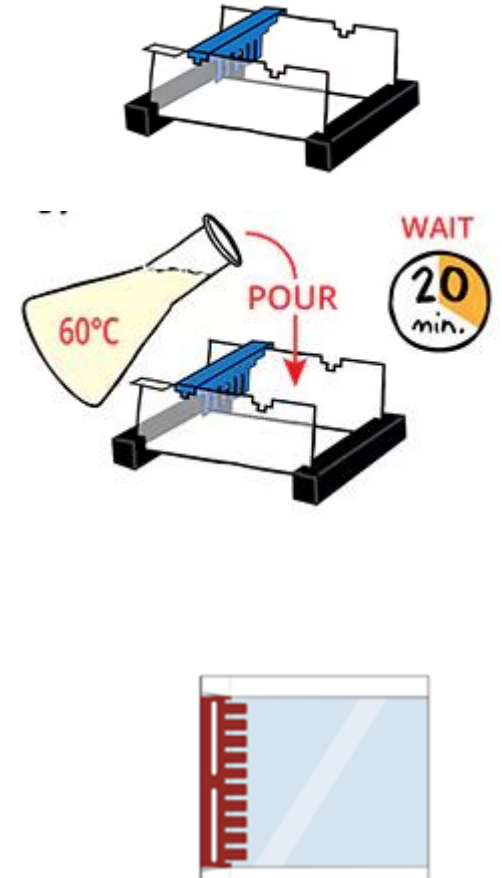


From Powder to Picture: A Step-by-Step Guide



Step 2: Pouring and Casting the Gel

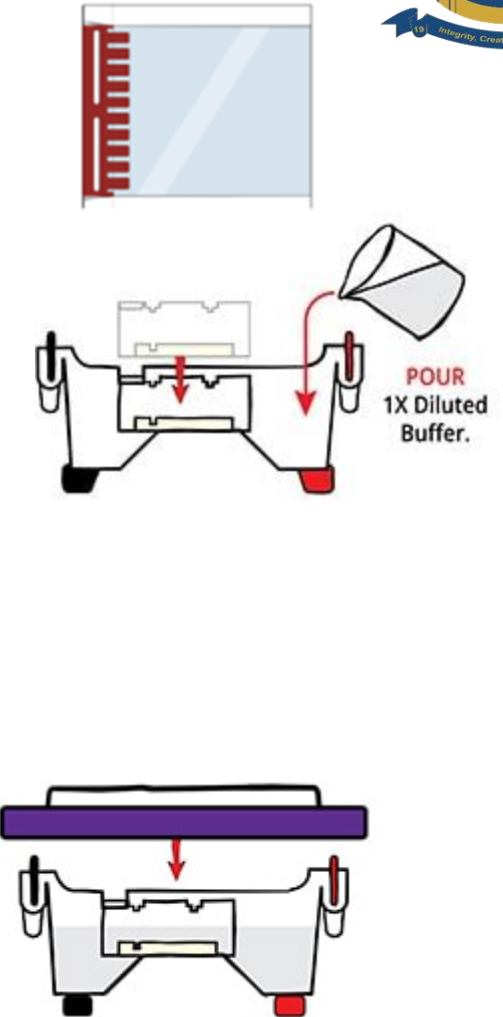
- ❖ Securely seal **both ends of the gel casting tray**. Place the **well-forming comb** into the notches at one end.
- ❖ Let the clear agarose solution cool to about **55-60°C**. It should be hot, but not boiling (cool enough to briefly touch the flask). Pouring when too hot can warp the plastic tray.
- ❖ Pour the **cooled agarose** into the **casting tray** slowly and **evenly to avoid creating air bubbles**.
- ❖ Let the gel sit undisturbed at room temperature for at **least 20-30** minutes until it is completely solid and has turned opaque.



From Powder to Picture: A Step-by-Step Guide

Step 3: Setting Up the Electrophoresis Chamber

- ❖ Place the solidified gel (still in its casting tray) into the electrophoresis chamber.
- ❖ Position the gel so the wells are at the negative (black) electrode end. Remember, DNA will "Run to Red" (the positive electrode).
- ❖ Pour running buffer into the chamber until the gel is completely submerged by a few millimeters. The buffer is needed to conduct the current.
- ❖ Carefully and slowly pull the comb straight up out of the gel to leave behind clean, well-defined wells.

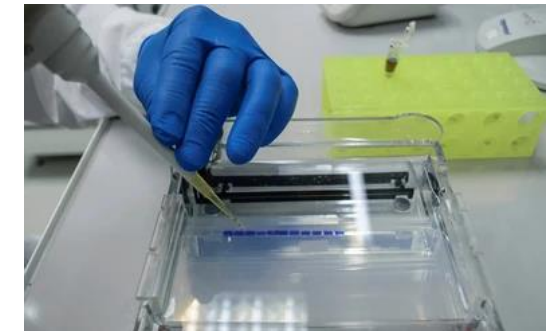


From Powder to Picture: A Step-by-Step Guide



Step 4: Loading the Samples

- ❖ Mix your DNA samples and DNA ladder with loading dye.
- ❖ Set your micropipette to the correct volume and draw up your first sample.
- ❖ **Technique is Key:**
 - ✓ Lower the pipette tip so it is just below the surface of the buffer and slightly inside the well opening. Do not puncture the bottom of the well.
 - ✓ Slowly and smoothly dispense the sample. You should see the dense, colored sample sink into the well.
- ❖ Use a fresh pipette tip for each new sample to prevent cross-contamination.

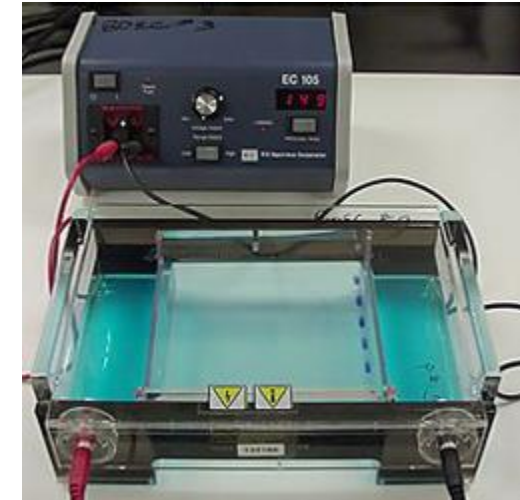


From Powder to Picture: A Step-by-Step Guide



Step 5: Running the Gel - The Migration

- ❖ Place the **lid** on the **electrophoresis chamber**. It should only fit one way.
- ❖ Connect the **electrical leads** to the power supply, matching the colors: **Red to Red (+)** and **Black to Black (-)**.
- ❖ Turn on the power supply to the recommended voltage.
- ❖ You should see **a stream of tiny bubbles** forming at the electrodes. This confirms that a current is flowing. You will see the colored tracking dyes slowly moving out of the wells and into the gel.

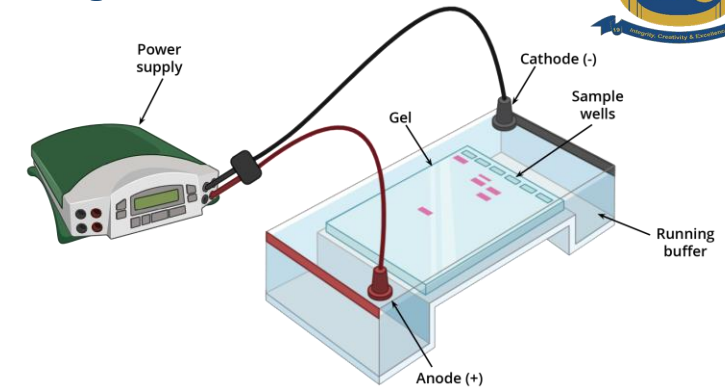


From Powder to Picture: A Step-by-Step Guide



Step 6: Visualizing the Results - The Reveal

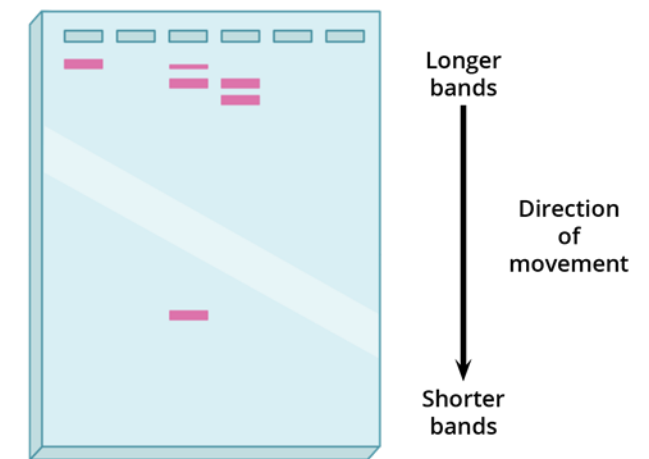
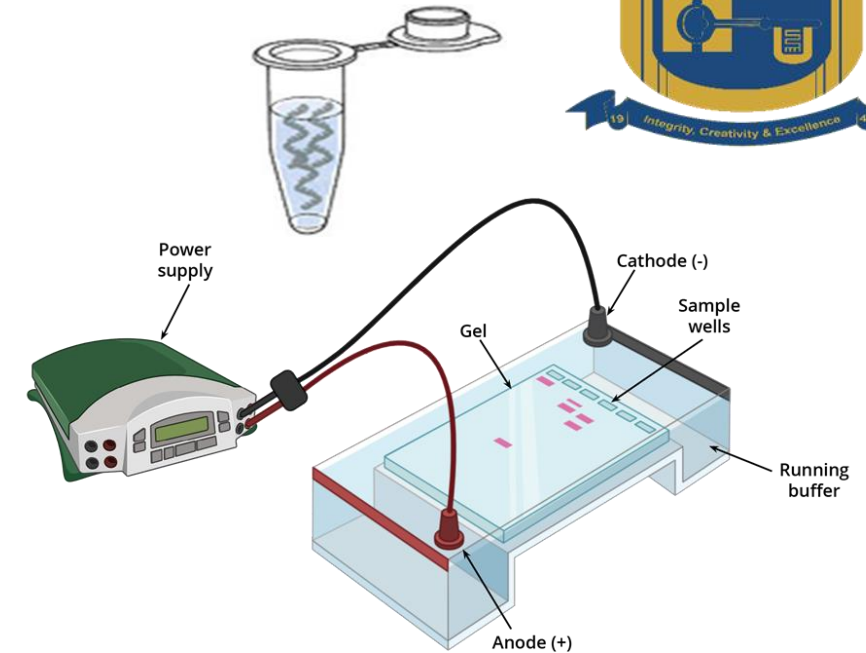
- ❖ When the tracking dye has migrated about **75% of the way down the gel**, turn off the power supply.
- ❖ **Carefully remove** the gel from the chamber. It is fragile and slippery.
- ❖ Place the gel onto the surface of a **UV transilluminator**.
- ❖ Lower the **UV-protective shield** before turning on the light.
- ❖ Turn on the **UV light to see the fluorescent DNA bands**.
Document the results by taking a picture for your analysis.



Putting It All Together: The Procedure

- ❖ **CAST the Gel:** Dissolve agarose in buffer, heat, pour, and let solidify.
- ❖ **SET UP:** Place gel in the chamber and submerge in buffer.
- ❖ **LOAD:** Mix DNA with loading dye. Pipette ladder and samples into wells.
- ❖ **RUN:** Connect leads (black-to-black, red-to-red) and apply voltage.
- ❖ **VISUALIZE:** View the gel on a UV trans illuminator.

sample prep -> power supply -> gel box -> final result



Thinking Like a Scientist: Analyzing a Gel



How to Read the Story on the Gel

- ❖ Identify the **DNA Ladder**. This is your ruler.
- ❖ Look at your **sample lane**. How many bands are there?
 - ✓ One band suggests a single, pure DNA product (like from PCR).
 - ✓ Multiple bands suggest a mixture of fragments (like from a digest).
 - ✓ A smear suggests DNA degradation.
- ❖ Estimate the **size of your band(s)** by **comparing** their position to the **closest bands on the ladder**.



Gel Electrophoresis Troubleshooting

❖ **Problem: "Smiling" bands (curved instead of flat).**

✓ **Cause:** Gel was run too fast (voltage too high), causing overheating.

❖ **Problem: No bands visible at all.**

✓ **Causes:** Forgot DNA stain? Leads connected backward? Not enough DNA?

❖ **Problem: Faint, blurry bands.**

✓ **Cause:** Low DNA concentration; diffusion from letting the gel sit too long after the run.

❖ **Problem: Bands are all clustered at the top.**

✓ **Cause:** Run time was too short.

Applications of Gel Electrophoresis

A Foundational Tool in Modern Biology

- ❖ **Forensics:** DNA fingerprinting.
- ❖ **Paternity Testing:** Comparing a child's DNA to potential fathers.
- ❖ **Medical Diagnostics:** Identifying genes related to diseases.
- ❖ **Molecular Cloning:** Verifying experimental results.
- ❖ **Evolutionary Biology:** Comparing DNA to determine species relationships.



Electrophoresis Safety & Waste

- ❖ **Electrical Hazard:** High voltage. Keep the lid on. No wet hands.
- ❖ **UV Light Hazard:** Can cause severe eye and skin damage.
Use the protective shield.
- ❖ **Chemical Hazard:** DNA stains can be mutagens. Always wear gloves.
- ❖ **Waste Disposal:** Gels and buffer with DNA stain go into designated hazardous waste.

Key Takeaways

- ❖ Gel electrophoresis separates DNA using an electric field.
- ❖ DNA has a negative charge and "runs to red."
- ❖ Separation is based on size, via molecular sieving.
- ❖ Small fragments move farther and faster than large fragments.
- ❖ Loading dye, DNA stain, and a DNA ladder are essential for visualization and analysis.

Practice Question

You perform **agarose gel electrophoresis** to analyze the **products of a restriction enzyme digest of a plasmid**. The digest is expected to produce **two DNA fragments**: **Fragment A (4,500 bp)** and **Fragment B (750 bp)**.



- After running the gel, which fragment (A or B) will be found closer to the loading wells at the top of the gel?
- Justify your answer by explaining the principle of molecular sieving and how it applies to the movement of these two fragments through the agarose matrix.



Questions?